sponsor's advertisement based on a random relevance score. Function score (`function_score`) queries help create a score based on user-defined functions, including random, script-based, or decay functions (Gauss, linear, etc.).

Before we start working with the `function_score` query, let's execute the query in the following listing. It is a straightforward `term` query that returns documents.

Listing 11.24 Term search with a standard `term` query

```
GET products/_search
{
  "query": {
    "term": {
      "product": {
        "value": "TV"
      }
    }
  }
}
```

The query doesn't do much other than search for a TV using a standard `term` query. The only point to note is the score of the top document returned by this query: 1.6376086.

Although this query is purposely simple, some queries require heavy processing to compute a relevance score. Maybe we aren't interested in fetching the score computed by Elasticsearch's Best Match 25 (BM25) relevance algorithm (or any custom algorithm) because we want to create a score based on our own requirements. In such a case, we can wrap a query in a `function_score` construct to generate a score based on user-defined functions.

Listing 11.25 A `term` search wrapped in a `function_score`

```
GET products/_search
{
  "query": {
    "function_score": {
      "query": {
        "term": {
          "product": "tv"
        }
      }
    }
  }
}
```

A function_score wraps a query to generate a user-defined score.

The `function_score` query expects a few attributes: a query, functions, how the score should be applied to the documents, and so on. Rather than learning about them in theory, we see these in the hands-on examples in this section.

User-defined functions let us modify and replace the score with our custom score. We can do so by plugging in a function that tweaks the score based on our